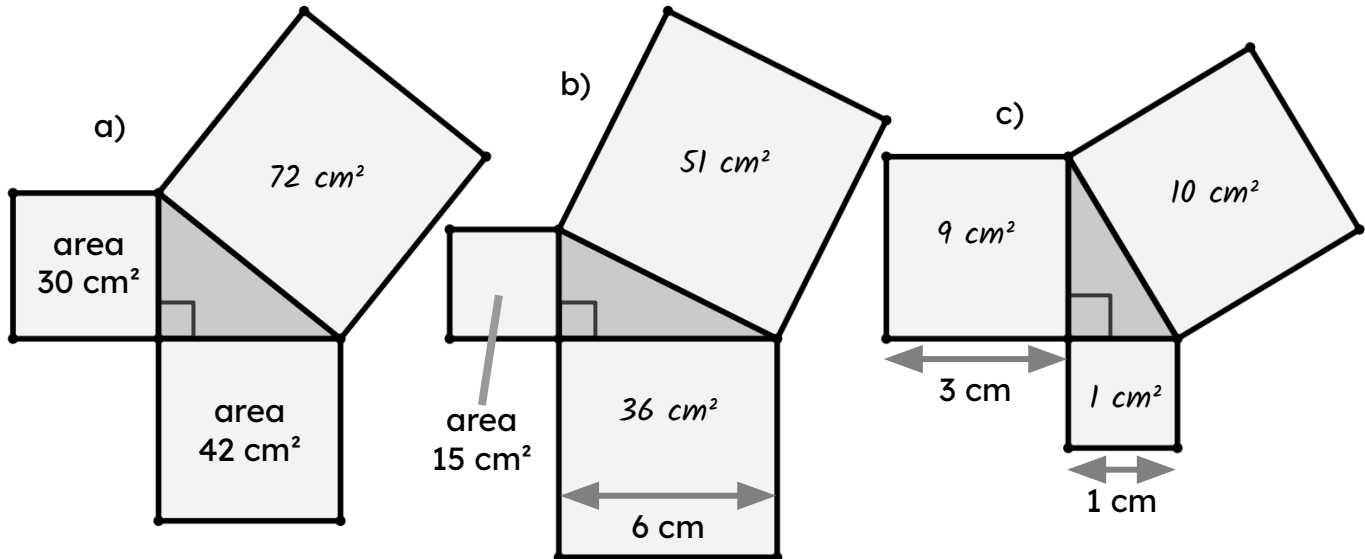




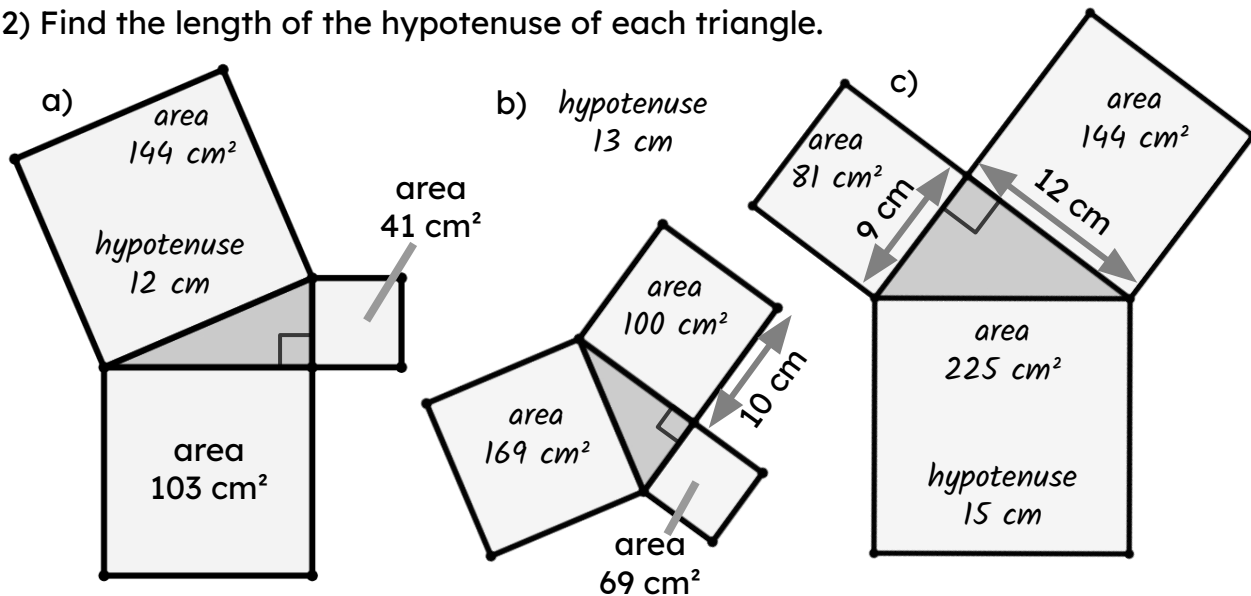
Length of the hypotenuse

Task A: Squares from the sides of a triangle

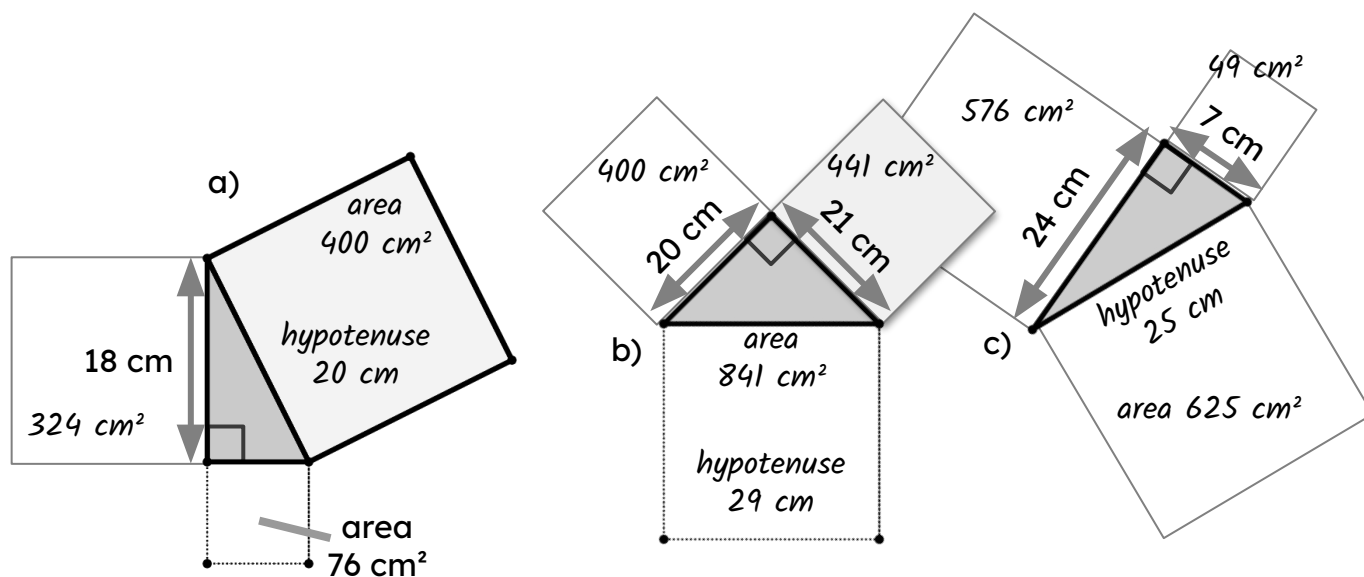
1) Find the area of each square.



2) Find the length of the hypotenuse of each triangle.

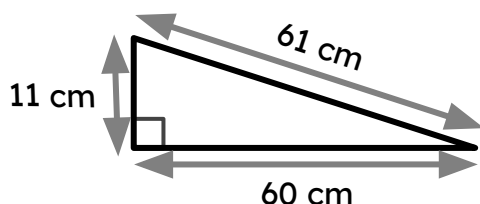


3) By drawing on any missing squares first and calculating their areas, find the length of the hypotenuse of each triangle.



Task B: Pythagoras' theorem to find the hypotenuse

1) This is a right-angled triangle.



$$61^2 = 60^2 + 11^2$$

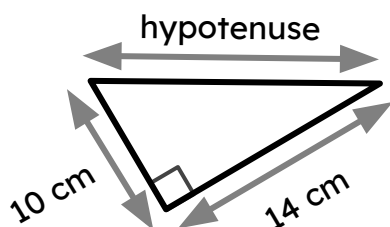
or

$$61^2 = 11^2 + 60^2$$

Pythagoras' theorem can be written as $c^2 = a^2 + b^2$, where c is the hypotenuse of the triangle.

Fill in the blanks to complete the Pythagoras' theorem equation for this triangle.

2) Show that the hypotenuse of this triangle is 17.2 cm (1 d.p.)

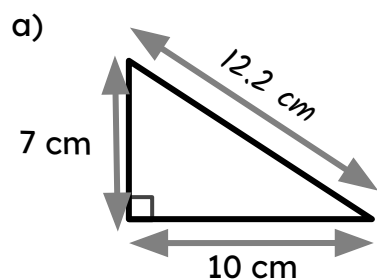


$$10^2 + 14^2 = 296$$

$$\sqrt{296} = 2\sqrt{74} = 17.20465...$$

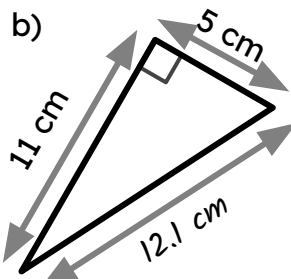
$$\text{hypotenuse} = 17.2 \text{ cm}$$

3) Match each triangle to its corresponding square of the hypotenuse and length of the hypotenuse.



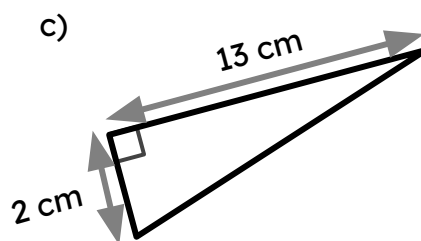
f 149 cm^2

g 12.2 cm



d 146 cm^2

i 12.1 cm



e 173 cm^2

h 13.2 cm

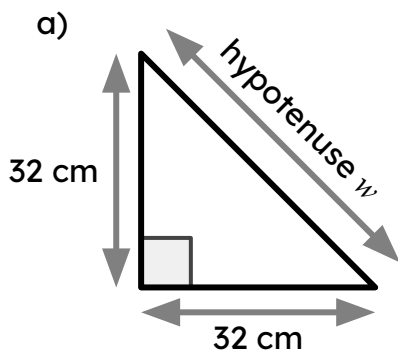
4) All these triangles have two lengths that sum to 64 cm.

Starting with the shortest, put the hypotenuses of the triangles in order of length.

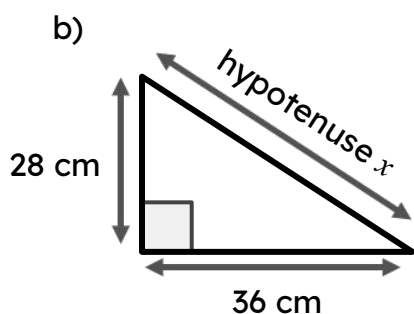
The correct order was

$w \rightarrow x \rightarrow y \rightarrow z$

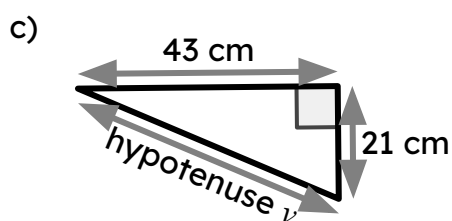
The bigger the difference between the two shorter side lengths, the larger the hypotenuse became.



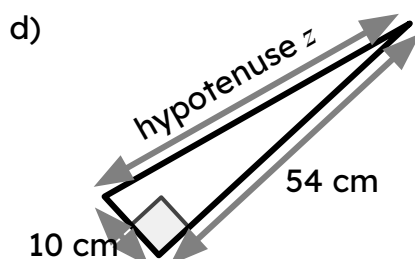
hypotenuse $w = 45.25 \text{ cm}$



hypotenuse $x = 45.61 \text{ cm}$



hypotenuse $y = 47.85 \text{ cm}$



hypotenuse $z = 54.92 \text{ cm}$

